Emergent Curvature from Accumulated Quantum Information

TZPID Companion Spine Series, Paper C22 of XXVII
The entanglement kernel behind the series' accumulation laws

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Abstract

This companion presents the field-theoretic framework in which spacetime curvature emerges from the temporal accumulation of quantum information: a non-local kernel $K(t,t') = \exp[-(t-t')/\tau_{\mathrm{dec}}]$ integrating stress-energy contributions into persistent geometric configurations. The source manuscript's stated thesis: "We present a comprehensive field-theoretic framework demonstrating that spacetime curvature emerges from the temporal accumulation of quantum information through entanglement dynamics. The central mechanism involves a non-local accumulation kernel K(t, t') = exp[(t t')/taudec] that integrates stress-energy contributions over time, generating persistent field configurations that manifest as geometric curvature." The paper organizes this thesis as a TZPID gold spine: a curated seven-node registry chain (ID3672, ID5818, ID9965, ID9967, ID9106, ID3673, ID9859) with typed proof-obligation roles, dictionary and encyclopedia anchors for every node, a declared dependency order, and stubs in the program's Isabelle/Wolfram verification pipeline. Within the arc this paper is the kernel's birthplace: the entanglement-dynamics reading of the accumulation machinery that Papers V, VI, and VII share.

1 Introduction

One mathematical object recurs in the core arc more than any other: the causal exponential kernel, $K(t,t') = \exp[-(t-t')/\tau_{\mathrm{dec}}]$, normalized and past-directed, integrating effective stress into accumulated curvature. Paper V used it for gravity, Paper VI for the residual, Paper VII for the threshold's build-up. This companion is the manuscript where that kernel was derived rather than deployed, and its derivation names the integrand's true currency: quantum information.

The framework's central mechanism is informational accumulation. Entanglement dynamics—the spreading and consolidation of quantum correlations among field degrees of freedom—deposits a record; the kernel integrates that record over time with decoherence timescale τ_{dec} setting the memory depth; and the integrated record manifests as persistent field configurations whose phenomenology is geometric curvature. Where Jacobson derived the Einstein equation from horizon entropy and Verlinde read Newton from information gradients, this framework is dynamical and local-in-history: curvature is not an equation of state but a running integral, and its source is not entropy in equilibrium but entanglement in transit.

The identification pays immediately in the core arc's terms. The kernel's normalization (Paper VI's certified guardrail) is conservation of informational weight; its causality is the arrow of decoherence; and τ_{dec}—the one free temporal parameter bracketed from two ends in Paper VI—acquires a microphysical meaning: the decoherence time of the accumulating correlations. The spine below anchors the framework's chain; refinement targets are the derivation's promotion to typed form and the matching of its information-theoretic τ_{dec} against the laboratory and cosmological brackets already in the registry.

2 The concept-anchored spine methodology

The companion series applies a uniform method, stated here so that the paper is self-contained. The source manuscript was published with its mathematics rendered as text rather than as machine-readable LaTeX, so its spine is *concept-anchored*: the thesis is taken from the manuscript's abstract, and the spine nodes are the canonical-registry entries most strongly matched to the manuscript's themes, selected by weighted term overlap with foundational nodes ranked first. Each node carries its registry equation, its dictionary definition, its encyclopedia interpretation, and a declared proof-obligation role (Core_Definition, Core_Axiom, or Derived_Theorem_Obligation). The resulting chain is a starting backbone for refinement—a typed scaffold onto which an exact, equation-by-equation crosswalk with the source manuscript can be progressively attached—rather than a claim of completed formalization.

Three properties make the backbone useful even before refinement completes. First, *addressability*: every claim in the companion paper resolves to a registry identifier whose equation, definition, and verification status are independently inspectable, so disagreement can be localized to a node rather than to the paper as a whole. Second, *pipeline coupling*: each spine is paired with an Isabelle focus-theory stub and a Wolfram check stub, so that as nodes are refined their obligations enter the same machine-checked pipeline that certifies the core series' guardrails. Third, *role discipline*: the obligation roles separate what the spine defines from what it asserts and what it must prove, which keeps the demarcation section of every companion paper honest by construction.

The dependency chain should accordingly be read as a proof-order proposal: upstream nodes supply the vocabulary and axioms that downstream nodes consume, and the refinement pass is free to reorder where the source manuscript's own logic demands it. Where a node's registry equation arrives with rendering artifacts inherited from the source extraction (a known cost of text-rendered mathematics), the equation is quoted as carried by the registry, with the dictionary and encyclopedia entries supplying the authoritative reading.

3 The registry chain

The curated chain, in proof order, is

$$\begin{gathered}\text{ID3672} \to \text{ID5818} \to \text{ID9965} \to \text{ID9967} \\ \to \text{ID9106} \to \text{ID3673} \to \text{ID9859}.\end{gathered}$$

We take the nodes in order; each subsection quotes the registry equation as carried by the canonical master, then gives the dictionary definition and the encyclopedia's interpretive reading.

3.1 ID3672: Magnetic Helicity Tensor Coupling (Core_Definition)

Registry equation:

```
$\mathcal{M} = T^*(SO(3) \times \mathbb{R}^3) \otimes \mathfrak{u}(1)_{\text{gauge}} \otimes
\cat{Info}_{\text{quantum}}$
```

Dictionary definition. Magnetic Helicity Tensor Coupling — M = T^*(SO(3) x R^3) (x) mathfraku(1)_gauge (x) catInfo_quantum

Encyclopedia reading. The entry preserves the equation exactly as extracted from ‘D:\00_EngineCSVemergent_curvature_the
for registry alignment and later derivation.

Computational guardrail: `emergent_curvature_t_ID3672_identity`.

Obligation reading. As a Core_Definition this node contributes vocabulary rather than assertion: downstream nodes may use its object freely, but nothing is yet claimed about the world. Its refinement burden is precision—the typed form must capture exactly the object the source manuscript deploys.

3.2 ID5818: Kinetic Term Relation (Core_Definition)

Registry equation:

```
$\mathcal{K}(t,t') = \exp[-(t-t')/\tau_{\text{decoherence}}]$
```

Dictionary definition. Kinetic Term Relation — K(t,t') = exp[-(t-t')/tau_decoherence]

Encyclopedia reading. The entry preserves the equation exactly as extracted from '00_EngineLemmaMakerPrismunified_field_theory_paper_centerpetal.tex' for registry alignment and later derivation.

Computational guardrail: `emergent_curvature_t_ID5818_identity`.

Obligation reading. This Core_Definition fixes an object the rest of the chain consumes. In proof order it must be discharged first; in refinement it is where a mismatch between the manuscript's usage and the registry's typing would first surface.

3.3 ID9965: Accumulation Kernel: delta S/delta g^munu (Core_Definition)

Registry equation:

```
\delta S/\delta g^{\mu\nu} = (c^4/16\pi G)G_{\mu\nu}
```

Dictionary definition. Accumulation Kernel: delta S/delta g^munu — delta S/delta g^munu = (c^4/16pi G)G_munu

Encyclopedia reading. Accumulation Kernel: delta S/delta g^munu is an algebraic definition, written delta S/delta g^munu = (c^4/16pi G)G_munu. It is an algebraic definition expressing the quantity directly in terms of the variables on its right-hand side. It is built from pi, mu, nu, c, G, G_, defining delta S/delta g^munu. Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: `emergent_curvature_t_ID9965_identity`.

Obligation reading. A definitional node: its function is to make later statements well-formed. The guardrail attached to it is correspondingly structural—an identity or well-formedness check rather than a physical claim.

3.4 ID9967: Accumulation Kernel: I(A:B) (Core_Definition)

Registry equation:

```
I(A:B) = S(A) + S(B) - S(AB)
```

Dictionary definition. Accumulation Kernel: I(A:B) — I(A:B) = S(A) + S(B) - S(AB)

Encyclopedia reading. Accumulation Kernel: I(A:B) is an algebraic definition, written I(A:B) = S(A) + S(B) - S(AB). It is an algebraic definition expressing the quantity directly in terms of the variables on its right-hand side. It is built from S, A, B, defining I(A:B). Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: `emergent_curvature_t_ID9967_identity`.

Obligation reading. As a Core_Definition this node contributes vocabulary rather than assertion: downstream nodes may use its object freely, but nothing is yet claimed about the world. Its refinement burden is precision—the typed form must capture exactly the object the source manuscript deploys.

3.5 ID9106: Definition of dcidlnmu (Core_Definition)

Registry equation:

```
dcidlnmu=fi(cj,dimeff)\frac{\text{d}c_i} {\text{d}\ln\mu}
```

Dictionary definition. Definition of dcidlnmu — dcidlnmu=fi(cj,dimeff)(dc_i)/(dlnmu)

Encyclopedia reading. Definition of dcidlnmu is an algebraic definition, written dcidlnmu=fi(cj,dimeff)(dc_i)/(dlnmu). It is an algebraic definition expressing the quantity directly in terms of the variables on its right-hand side. It is built from mu, f, c, j, m, c_i, defining dcidlnmu. Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: `emergent_curvature_t_ID9106_identity`.

Obligation reading. This Core_Definition fixes an object the rest of the chain consumes. In proof order it must be discharged first; in refinement it is where a mismatch between the manuscript's usage and the registry's typing would first surface.

3.6 ID3673: Grav Quantum Charge Functional (Derived_Theorem_Obligation)

Registry equation:

```
$\hat{Q}_{\text{grav}} = (\hbar c^3/8\pi G)\int_{-\infty}^t [\partial I/\partial\tau - \partial
S/\partial\tau]\exp[-(t-\tau)/\tau_{\tex t{dec}}]d\tau$
```

Dictionary definition. Grav Quantum Charge Functional — Q_grav = (c^3/8pi G)integral_-infinity^t [d I/dtau - d S/dtau]exp[-(t-tau)/tau_dec]dtau

Encyclopedia reading. The entry preserves the equation exactly as extracted from 'D:\00_EngineCSVemergent_curvature_the for registry alignment and later derivation.

Computational guardrail: `emergent_curvature_t_ID3673_identity`.

Obligation reading. This node carries proof debt by declaration—Derived_Theorem_Obligation—and is therefore where the refinement pass should concentrate first: a discharged derivation here strengthens every downstream node simultaneously.

3.7 ID9859: Stress–Energy Tensor Relation (Derived_Theorem_Obligation)

Registry equation:

```
\int_{\Sigma} \sqrt{|g|}, R^{\mu\nu} T_{\mu\nu}, d^4x
```

Dictionary definition. Stress–Energy Tensor Relation — integral_Sigma sqrt(|g|), R^munu T_munu, d^4x

Encyclopedia reading. Stress–Energy Tensor Relation is an integral expression, written integral_Sigma sqrt(|g|), R^munu T_munu, d^4x. It is obtained by integrating over a domain, accumulating contributions across the region of interest. It is built from Sigma, mu, nu, g, R, T_. Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: `emergent_curvature_t_ID9859_structure`.

Obligation reading. As a Derived_Theorem_Obligation this node owes a proof: the registry asserts that it follows from the chain's upstream vocabulary and axioms, and the Isabelle focus stub holds the slot where that derivation will be discharged.

4 Dependency structure and obligations

Table 1 summarizes the spine's obligation ledger. The chain's proof order runs from the definitional nodes (vocabulary first) through the derived obligations to the spine's closure; the refinement pass may interleave additional registry nodes where the source manuscript's own derivations require them, in which case the table grows monotonically—existing rows are never weakened, only joined.

Table 1: Obligation ledger for the emergent curvature theory spine.

ID	Obligation role	Wolfram check
ID3672	Core_Definition	`emergent_curvature_t_ID3672_identity`
ID5818	Core_Definition	`emergent_curvature_t_ID5818_identity`
ID9965	Core_Definition	`emergent_curvature_t_ID9965_identity`
ID9967	Core_Definition	`emergent_curvature_t_ID9967_identity`
ID9106	Core_Definition	`emergent_curvature_t_ID9106_identity`
ID3673	Derived_Theorem_Obligation	`emergent_curvature_t_ID3673_identity`
ID9859	Derived_Theorem_Obligation	`emergent_curvature_t_ID9859_structure`

5 Crosswalk to the core series and companion corpus

Because all spines draw on one canonical registry, node sharing is measurable and meaningful: a shared identifier means two papers consume literally the same typed object, so verification work transfers between them without translation. Of this spine's seven nodes, 0 appear in core-series spines and 2 recur elsewhere in the companion corpus. Table 2 records the census.

Table 2: Node-sharing census for this spine.

ID	Core-series appearances	Companion-corpus appearances
ID3672	—	Universal Field Theory Rendered
ID5818	—	—
ID9965	—	—
ID9967	—	—
ID9106	—	—
ID3673	—	—
ID9859	—	trawin enlil protocol

Two readings follow. Nodes shared with the core series are this spine's strongest members: their guardrails are already certified in the core pipeline, and the present paper inherits that status. Nodes unique to this spine are its distinctive contribution—and its refinement priority, since no other paper will discharge them. Nodes recurring across companions mark the corpus's internal seams: the same object consumed by several manuscripts, whose refinements must agree by the duplicate discipline documented in the Einstein Focus companion.

6 Position in the TZPID arc

This companion stands behind every accumulation law in the core series: it is the microphysics of the kernel that Papers V–VII share, and it joins C25 (information theory of TZPs) to give the arc's geometry an informational substrate. Against the thermodynamic emergent-gravity programs, it occupies the position Paper

V staked structurally: history-dependent, conservative by certificate, and falsifiable through the kernel's measurable timescale.

7 Verification pathway

The spine enters the program's two-engine verification pipeline. The Isabelle focus theory `TZPID_Spine_emergent_curvat` types the seven obligations over the registry's declared vocabulary, in the dependency order of Section 3; the Wolfram check stub `emergent_curvature_theory_checks.wl` carries the per-node computational guardrails listed in Table 1. As with the core series, the division of labor is deliberate: the theorem prover certifies structure (types, roles, dependency soundness), while the computational engine certifies arithmetic and algebraic identities node by node. A check name of the form `..._identity` denotes a per-node identity guardrail generated from the registry equation; refined checks replace these as the equation-level crosswalk matures.

Because the companion spines share the registry with the core series, verification is cumulative: any node here that also appears in a core spine (the chain tables record several) inherits the core guardrail's status, and any refinement proved here propagates back. The companion series thus functions as the proof pipeline's breadth dimension—161 distinct registry nodes across the 27 companions—complementing the core series' depth.

The pipeline's two-engine division also fixes what failure means at this layer. A failed Wolfram guardrail localizes an arithmetic or algebraic defect in one node's registered equation: the repair is editorial (correct the registration against the source) or substantive (the source itself errs), and the obligations ledger records which. A failed Isabelle discharge, by contrast, indicts the chain: a type mismatch or unprovable dependency means the proof-order proposal of Section
refsec:chain is wrong as stated, and the refinement pass must re-curate. Keeping these failure modes separate is what the stub architecture buys: every companion enters the pipeline with its failure semantics declared in advance, which is the practical meaning of being peer-reviewable by machine.

8 Refinement worksheet

The concept-anchored backbone converts into an equation-level formalization through a bounded, node-by-node worksheet. For each node the worksheet item below states the concrete refinement act; completing all seven closes the spine's first formalization pass.

1. **ID3672** (`emergent_curvature_t_ID3672_identity`): exhibit the source manuscript's working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.

2. **ID5818** (`emergent_curvature_t_ID5818_identity`): exhibit the source manuscript's working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.

3. **ID9965** (`emergent_curvature_t_ID9965_identity`): exhibit the source manuscript's working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.

4. **ID9967** (`emergent_curvature_t_ID9967_identity`): exhibit the source manuscript's working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.

5. **ID9106** (`emergent_curvature_t_ID9106_identity`): exhibit the source manuscript's working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.

6. **ID3673** (`emergent_curvature_t_ID3673_identity`): reconstruct the source manuscript's derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.

7. **ID9859** (`emergent_curvature_t_ID9859_structure`): reconstruct the source manuscript's derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.

The worksheet's order matters: definitional items unblock derived ones, and a failure at any item halts propagation rather than silently weakening downstream claims. Completed items should update the obligations CSV in place, so that the spine's public ledger always reflects its true refinement state—the same live-ledger discipline the core series applies to its guardrails.

9 Demarcation and refinement targets

As throughout the program, the demarcation line is drawn explicitly. The established layer comprises the standard mathematics and physics that the chain's nodes quote—definitions, identities, and independently citable results—together with whatever the computational guardrails certify. The framework layer comprises the source manuscript's interpretive claims and the spine's structural assertion that its nodes form the stated dependency chain. That curvature tracks integrated entanglement is the framework's central conjecture; its near-term audit is consistency—one τ_{dec} serving laboratory decoherence, Paper VI's residual brackets, and the cosmological growth history simultaneously.

Refinement targets, in pipeline order: (i) replace the concept-anchored node selection with an equation-level crosswalk against the source manuscript; (ii) promote each `..._identity` guardrail to a content-bearing check; (iii) discharge the typed obligations in the Isabelle focus theory against the program's shared vocabulary; and (iv) record any node whose refinement fails, since a failed node localizes exactly where the source manuscript and the canonical registry disagree—the information a refinement pass exists to produce.

10 Conclusion

Emergent curvature theory gives the series' favorite integral its origin story: the kernel as decoherence-weighted memory, stress as informational deposit, geometry as the running total. Its refinement is the τ_{dec} consistency audit across every bracket the registry holds. If one timescale serves all three regimes, the accumulation principle acquires a microphysical engine; if not, the kernel's unity across Papers V–VII was bookkeeping rather than physics—either outcome instructs the arc.

References

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